



DataSentinel

USER & INTEGRATION GUIDE

Protect personal data. Prove compliance. Stay in control.

The India-first, DPDP-native platform for discovering, governing and enforcing the protection of personal data across your cloud, databases, APIs and AI systems.

DPDP Act 2023 ready

PII discovery & classification

Inline enforcement gateway

AI / LLM data guard

Data-principal rights

Audit-ready evidence

Document: Product & Integration Handbook

Audience: Security, Compliance & Platform teams

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00 About this guide

Everything you need to deploy DataSentinel, connect your data, and run a continuous personal-data protection program.

This handbook is written for the people who will operate DataSentinel day to day — security engineers, data-protection officers (DPOs), compliance leads and platform/DevOps teams. It explains what the product does, how each capability maps to a real compliance or security outcome, and exactly how to integrate it into your existing infrastructure.



What it does

Discovers personal data, enforces policy on data in motion, and manages your DPDP obligations.



How to integrate

Connect cloud assets and databases, and route sensitive traffic through the enforcement gateway.



How to monitor

Use the dashboard, live event feed, alerts and reports to watch your posture continuously.

Table of contents

01 Introduction & overview

02 Platform architecture

03 Core capabilities at a glance

04 Getting started

05 Connecting & monitoring assets

06 The enforcement gateway

07 AI / LLM data guard

08 Policies & compliance packs

09 Data-principal rights (DSRs)

10 Consent management

11 Reports & evidence

12 Alerts, webhooks & integrations

13 API & automation

14 Monitoring & observability

15 Security & multi-tenancy

16 Deployment options

17 Best practices

18 Troubleshooting & FAQ

A Appendix — PII types

B Appendix — Enforcement actions

C Appendix — Glossary & support

01 Introduction & overview

DataSentinel is a data governance and sovereignty platform built specifically for the Indian Digital Personal Data Protection (DPDP) Act, 2023.

Modern organisations spread personal data across object storage, managed databases, internal APIs and — increasingly — third-party AI services. Most teams have no continuous visibility into *where* that data lives, *where it flows*, or *whether it is leaking* to places it should not. DataSentinel closes that gap with three coordinated functions:



Discover

Scan cloud and database assets to find and classify personal data — with recognisers tuned for Indian identifiers.



Enforce

Inspect live request/response traffic through an inline proxy and mask, redact, block or tokenize sensitive fields in real time.



Govern

Manage rights requests, consent, policies and audit-ready reporting against your DPDP obligations.

Who it is for

- **Data-protection officers & compliance teams** — demonstrate DPDP compliance with living evidence, not spreadsheets.
- **Security engineers** — stop PII from leaking into unauthorised APIs, storage and AI systems.
- **Platform / DevOps teams** — deploy lightweight services into existing Kubernetes or Docker estates with minimal friction.
- **Leadership** — a single compliance score and risk posture across the whole organisation.

Why DataSentinel

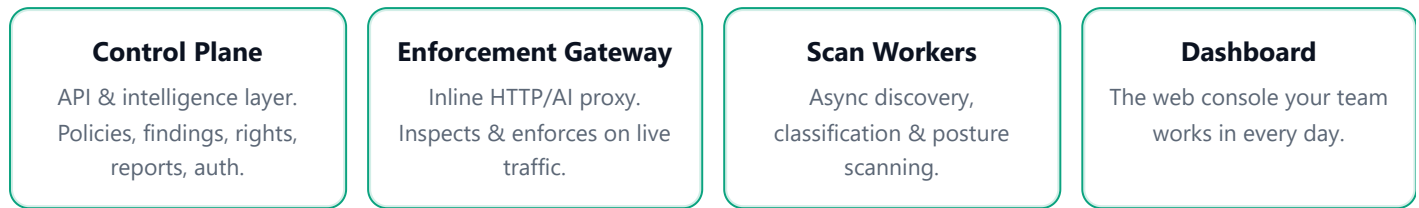
Challenge	How DataSentinel helps
"We don't know where personal data is."	Automated discovery & classification across S3, RDS and PostgreSQL with column- and object-level findings.
"Sensitive data leaks into external APIs and LLMs."	An inline enforcement gateway that masks, blocks or tokenizes PII in real time, including dedicated AI/LLM protection.
"We can't respond to data-principal requests in time."	Built-in DSR workflow with 90-day deadline tracking and cross-asset data search.
"Audits are painful and manual."	One-click DPDP, DPIA and audit-evidence reports backed by an immutable event trail.
"Our data must stay in India."	Data-sovereignty controls, India-region defaults and cross-border transfer policies.

OUTCOME

Within days of connecting your first asset you have a measured compliance posture, a prioritised list of exposures, and live enforcement preventing new leaks — without re-architecting your applications.

02 Platform architecture

Four lightweight, independently deployable services backed by standard data stores.



▲ All four services share encrypted state and a common tenant model ▲

The services

- **Control Plane** — the brain. Exposes the REST API, evaluates policies, stores findings, drives reports and orchestrates scans.
- **Enforcement Gateway** — a high-performance reverse/egress proxy that inspects request and response bodies and applies policy actions inline.
- **Scan Workers** — background workers that connect to your assets, sample data, classify PII and check cloud posture.
- **Dashboard** — a modern web UI for analysts, DPOs and admins.

The data stores

- **PostgreSQL 16** — application state (assets, policies, findings, rights, consent), isolated per tenant with row-level security.
- **Redis 7** — policy cache, the tokenization vault and the worker task queue.
- **ClickHouse 24** — high-volume audit logs and gateway events, retained for years for evidence.

You can run these as managed services (Amazon RDS, ElastiCache, ClickHouse Cloud) or self-host them.

How data moves

1. **Discovery:** Workers connect to a registered asset, sample records, and write classified findings to the control plane.
2. **Enforcement:** Application traffic routed through the gateway is inspected; matched policies mask/redact/block/tokenize PII before it leaves your estate. Every decision is logged.
3. **Governance:** Findings, gateway events, rights requests and consent records roll up into the dashboard, alerts and compliance reports.

DEPLOYMENT FLEXIBILITY

Use the managed SaaS, or run the entire platform inside your own VPC/Kubernetes cluster for full data residency. See **Section 16 — Deployment options**.

03 Core capabilities at a glance

A single platform that spans discovery, real-time enforcement and compliance operations.



PII discovery & classification

Find personal data in S3, RDS and PostgreSQL using Indian-tuned recognisers (Aadhaar, PAN, UPI, GSTIN and more) with checksum validation.



Inline enforcement gateway

Mask, redact, block, tokenize, encrypt, hash or alert on PII in live request/response traffic.



AI / LLM data guard

Strip personal data from prompts to OpenAI, Anthropic, Google & others before it ever leaves your network.



Security posture

Detect risky misconfigurations — public buckets, missing encryption, weak transport security.



Data-principal rights

Track access, correction, erasure, portability & nomination requests against the 90-day deadline.



Consent management

Record, summarise and withdraw consent by purpose with a full history.



Policies & templates

Ready-made DPDP, RBI, IRDAI, AI-guard and security policy packs you can apply in one click.



Reports & evidence

Generate DPDP compliance, DPIA, asset-inventory and audit-evidence reports on demand.



Alerts & webhooks

Route incidents to Slack, PagerDuty, JIRA, email or any HTTP endpoint.



Data-flow mapping

Visualise where PII flows to — and approve or flag each destination.



Live dashboard

A real-time compliance score, risk trends and a live gateway event feed.



Enterprise security

Per-tenant encryption, row-level isolation, RBAC and full audit logging.

04 Getting started

From sign-in to your first findings in five steps.

1 Sign in & create your workspace

Register your organisation (which becomes your isolated *tenant*) and your first administrator account. Invite teammates from **Settings** → **Team** and assign each a role (owner, admin, analyst or viewer).

2 Connect your first asset

Go to **Assets** → **Connect asset**, choose a type (e.g. Amazon S3 or PostgreSQL), and provide read-only connection details. Use **Test connection** to confirm credentials before saving.

3 Run a scan

Open the asset and click **Run scan**. Workers sample records, classify PII and write findings. Incremental scans also run automatically on a schedule.

4 Review findings

Visit **Findings** to see what was discovered, ranked by severity. Each finding shows the PII types, location and a remediation hint. Filter by severity, type or asset.

5 Apply a policy & turn on enforcement

From **Policies** → **Templates**, apply the DPDP starter pack, then deploy the gateway (Section 6) to begin masking or blocking PII on live traffic.

THE DASHBOARD AT A GLANCE

Your home screen shows a **compliance score**, total PII records and assets, open and critical findings, unacknowledged alerts, overdue rights requests, a PII-type breakdown, a 30-day findings trend, your top-risk assets and a live data-flow map.

05 Connecting & monitoring assets

Assets are the cloud and database resources DataSentinel scans for personal data.

Supported asset types

Type	What it scans	Status
Amazon S3 <code>s3_bucket</code>	Objects (JSON, CSV, NDJSON, text/log) sampled for PII; bucket security posture.	Available
PostgreSQL <code>postgresql</code>	Tables & columns sampled for PII using column-name heuristics first.	Available
Amazon RDS <code>rds_instance</code>	PostgreSQL-compatible RDS instances; data & transport posture.	Available
Google Cloud Storage / Azure Blob	Object storage discovery.	Roadmap
API & LLM endpoints	Governed live via the enforcement gateway (Sections 6–7).	Available

Connection configuration

Credentials are stored **encrypted** (AES-256-GCM with a per-tenant key) and are only ever decrypted in-memory by a worker during a scan. Always use a **read-only** principal.

Amazon S3

```
# connection_config
{
  "bucket_name": "acme-customer-data",
  "region":      "ap-south-1",
  "prefix":      "exports/",
  "role_arn":    "arn:aws:iam:: ... :role/ds-scan"
}
```

Grant the role `s3:ListBucket`, `s3:GetObject` and the read-only bucket posture permissions.

PostgreSQL / RDS

```
# connection_config
{
  "host":      "db.internal:5432",
  "database":  "app",
  "username":  "ds_readonly",
  "password":  ".....",
  "schema":   "public",
  "ssl_mode":  "require"
}
```

Create a role with `SELECT` on the schemas you want scanned.

Scanning & monitoring

- **Manual scans** — run on demand from any asset's page.
- **Scheduled scans** — assets not scanned in the last 24 hours are picked up automatically.
- **Posture checks** — run periodically to surface misconfigurations as `misconfiguration` findings (e.g. public bucket, no default encryption, versioning disabled, TLS not enforced).
- **Per-asset view** — drill into findings, scan history and detected data flows for each asset.
- **Risk score** — every asset receives a 0–100 score derived from finding severity and PII volume.

PRIVACY BY DESIGN

Scans never store raw personal data. Findings keep only metadata — PII type, location, counts and safely-masked samples — so DataSentinel itself never becomes a new copy of your sensitive data.

06 The enforcement gateway

Stop personal data from leaking — in real time — by routing traffic through the gateway.

The gateway is a transparent HTTP proxy you place in front of selected API endpoints, or use as an *egress* proxy for outbound calls to external services. It parses request and response bodies, runs the PII detection engine, and applies your policy actions **before** data leaves your estate. Every decision is logged for evidence; typical added latency is only a few milliseconds.

How to route traffic

Point your client (or egress configuration) at the gateway and tell it where the request should ultimately go using request headers:

Header	Required	Purpose
X-Upstream-URL	Yes	The real destination the gateway should forward to.
X-Tenant-ID	Yes*	Identifies your tenant (or derived from a verified JWT).
X-API-Key	Yes*	Service authentication for the gateway.
Authorization: Bearer	Optional	RS256 JWT — verified, with the tenant taken from the token.

*Provide either an API key + tenant header, or a verified Bearer JWT.

Example — proxy an outbound API call

```
curl https://gateway.acme.com/v1/customers \  
-H "X-API-Key: <service-key>" \  
-H "X-Tenant-ID: 7f3c..." \  
-H "X-Upstream-URL: https://partner-api.example.com/v1/customers" \  
-H "Content-Type: application/json" \  
-d '{ "pan": "ABCDE1234F", "phone": "+91 98xxxxxx10" }' \  
# → PII is masked/blocked per policy before reaching the partner API
```

Enforcement actions

Each gateway rule matches a route, direction (request/response) and PII types, then applies one action:

Action	Effect
mask	Replace characters while keeping format (e.g. XXXX-XXXX-1234).
redact	Remove the field from the payload entirely.
block	Reject the request with HTTP 403 and log the violation.
tokenize	Swap the value for a reversible, format-preserving token from the per-tenant vault.
encrypt	AES-256-GCM encrypt the field with the tenant key.
hash	One-way SHA-256 pseudonymisation.
alert	Allow the traffic but raise an alert.

BUILT-IN SAFETY

The gateway refuses to proxy to cloud metadata and link-local endpoints (SSRF protection), fails open if policy can't be loaded so legitimate traffic is never dropped by accident, and exposes `/healthz`, `/readyz` and Prometheus `/metrics`.

07 AI / LLM data guard

Adopt AI safely — without sending customer PII to third-party models.

When traffic targets a known LLM provider, the gateway switches into **LLM mode**. It understands chat/completion request formats, scans every prompt segment for personal data, and sanitises it before forwarding. It also inspects the model's response for unexpected PII leakage.

Recognised providers

- OpenAI — `api.openai.com`
- Anthropic — `api.anthropic.com`
- Google — `generativelanguage.googleapis.com`
- Azure OpenAI & AWS Bedrock

What it does

- Redacts/masks PII in prompts pre-flight.
- Scans responses for leaked personal data.
- Logs sanitised + hashed originals for audit.
- Can block requests carrying sensitive financial data.

Example — guard an OpenAI call

```
curl https://gateway.acme.com/v1/chat/completions \
-H "X-API-Key: <service-key>" \
-H "X-Tenant-ID: 7f3c..." \
-H "X-Upstream-URL: https://api.openai.com/v1/chat/completions" \
-H "Content-Type: application/json" \
-d '{ "model": "gpt-4o", "messages": [
  { "role": "user", "content": "Summarise: Aadhaar 1234 5678 9012 ..." }
] }'
```

→ Aadhaar is masked before the prompt reaches OpenAI

READY-MADE POLICY

Apply the **AI / LLM Guard pack** (Section 8) to redact all PII from prompts, alert on PII in responses, and block prompts containing sensitive financial data — no custom rules required.

08 Policies & compliance packs

Express your data-protection rules once, then enforce them everywhere.

A **policy** is a set of rules that decide what happens when personal data is detected. Each rule combines conditions (PII type, direction, destination) with an action (mask, block, tokenize, ...) and an enforcement mode:

- **Alert-only** — observe and report without changing traffic (great for rollout).
- **Enforce** — actively apply the action.
- **Audit-only** — record for evidence.

Build policies visually in the dashboard, see a live preview of the underlying rule definition, and version every change with one-click rollback.

Built-in template packs

DPDP compliance pack

- Mask all Indian PII in responses to unapproved destinations
- Block transfers outside approved countries
- Alert on personal data in LLM prompts
- Redact PII from application logs
- 90-day rights-request deadline alerts
- Breach-indicator & mass-export detection
- Children's-data consent guardrails
- Retention-period enforcement

RBI data-localization pack

- Keep payment data on Indian infrastructure
- Alert on transaction data to non-approved endpoints
- Mask card numbers (PCI-style, keep last 4)
- Tokenize account numbers in API traffic

IRDAI pack

- Alert on policyholder data in cross-border transfers
- Mask health-data fields on non-clinical paths

AI / LLM guard pack

- Redact PII from prompts to external LLMs
- Alert when responses contain new personal data
- Block prompts with sensitive financial data
- Watermark responses for leak tracing

General security pack

- Alert on bulk exports (>1000 records)
- Block unauthorised IP ranges
- Alert on PII access outside business hours
- Detect credentials/secrets in traffic
- Rate-limit PII-bearing endpoints

APPLY IN ONE CLICK

From **Policies** → **Templates**, choose a pack and apply it. Start in alert-only mode, review the gateway live feed, then switch to enforce when you're confident.

09 Data-principal rights (DSRs)

Meet your DPDP obligation to respond to individuals within 90 days.

When an individual (a “data principal”) exercises their rights, log the request in **Rights**. DataSentinel tracks it through its lifecycle and automatically computes the statutory deadline.

Request type	What DataSentinel does
Access	Searches every connected asset for the principal’s records and compiles a structured export.
Correction	Locates the records and flags exact asset/table/location for correction.
Erasure	Identifies all locations of the data and queues erasure (final deletion stays human-approved for safety).
Portability	Extracts the principal’s data in a machine-readable format.
Nomination	Records a nominee for the data principal.

Workflow

1. **Received** — request logged; 90-day due date set automatically.
2. **Assign** — route to an owner on your team.
3. **Search** — trigger a cross-asset search to locate the individual’s data.
4. **Complete or reject** — record the response data, or a documented rejection reason.

NEVER MISS A DEADLINE

DataSentinel raises **high-severity** alerts as deadlines approach and **critical** alerts when a request becomes overdue. The dashboard surfaces overdue counts and an upcoming-deadline calendar.

10 Consent management

Capture and prove lawful basis for processing, by purpose.

Record each consent event — who consented, for what purpose, when, through which mechanism, and the notice version they saw. DataSentinel maintains the full history, including withdrawals, so you can demonstrate lawful basis at any moment.

What you can do

- Record consent via the dashboard or API/SDK.
- Look up all consents for a single data principal.
- Record a withdrawal (with timestamp).
- Bulk-import existing consent from CSV.

What you can see

- Aggregate consent given vs. withdrawn.
- Breakdown by processing purpose.
- Consent mechanism (form, API, SDK, import).
- Per-principal consent timeline.

WHY IT MATTERS

Consent records feed directly into your DPDP and DPIA reports, evidencing that processing has a valid, documented basis — and that withdrawals are honoured.

11 Reports & evidence

Turn your live posture into audit-ready documents on demand.

Generate reports from **Reports** → **Generate**. Each is assembled from your current data and made available for download.

Report	Use it for
DPDP compliance summary	Section-by-section status of your DPDP obligations.
DPIA	Data-protection impact assessment with assets, findings and consent.
Asset inventory	A full register of connected data assets and their risk.
Executive summary	Leadership-level metrics and risk posture.
Incident report	Findings plus recent scan history for an event.
Audit-evidence pack	Scans, findings, rights and consent bundled for auditors.

BACKED BY AN IMMUTABLE TRAIL

Reports draw on an event store that retains gateway decisions and audit logs for years, so the evidence behind every number is always available.

12 Alerts, webhooks & integrations

Get the right signal to the right team, in the tools you already use.

DataSentinel raises alerts for policy violations, detected breaches, scan anomalies, rights-request deadlines, retention events and cross-border transfers. Acknowledge them in-app individually or in bulk, and fan them out to your operational tooling.



Slack

Post incidents to a channel.



PagerDuty

Page on-call for criticals.



JIRA

Open tickets automatically.



Email

Notify recipient lists.



Generic webhook

Signed HTTP POST to any URL.



Preferences

Tune channels & thresholds.

Configure all of this under **Settings** → **Integrations**: add a webhook, choose which event types it receives, send a test event, and copy the signing secret (shown once) so you can verify authenticity on your side.

SECURE BY DEFAULT

Webhook payloads are signed. Always verify the signature using your stored secret before acting on an event.

13 API & automation

Everything in the dashboard is available through a clean REST API.

Automate onboarding, scanning, reporting and rights workflows. The API base path is `/api/v1`. Authenticate with a user's Bearer JWT or, for machine-to-machine use, an **API key**.

Create & use an API key

Create keys under **Settings** → **API Keys** with scoped permissions (read, write, gateway, admin). The full key is shown only once — store it securely.

```
# List assets using an API key
curl https://app.datasentinel.io/api/v1/assets \
  -H "X-API-Key: ds_live_*****"

# Trigger a scan
curl -X POST https://app.datasentinel.io/api/v1/assets/<id>/scan \
  -H "X-API-Key: ds_live_*****" \
  -d '{ "scan_type": "full" }'
```

Frequently used endpoints

Area	Endpoints
Assets	GET/POST <code>/assets</code> , <code>/assets/:id/scan</code> , <code>/assets/:id/findings</code>
Findings	GET <code>/findings</code> , <code>/findings/summary</code> , <code>/findings/:id/resolve</code>
Policies	GET/POST <code>/policies</code> , <code>/policies/templates</code> , <code>/:id/activate</code>
Gateway	<code>/gateway/rules</code> , <code>/gateway/events</code> , <code>/gateway/stats</code> , <code>/gateway/data-flows</code>
Rights	GET/POST <code>/rights-requests</code> , <code>/:id/assign</code> , <code>/:id/complete</code>
Reports	GET/POST <code>/reports</code> , <code>/reports/templates</code>
Consent	<code>/consent/record</code> , <code>/consent/summary</code> , <code>/consent/withdraw/:id</code>

All responses use a consistent envelope: `{ data, meta, error, request_id }` with pagination metadata where relevant.

14 Monitoring & observability

Watch your data-protection posture continuously.

Score

LIVE COMPLIANCE

Feed

REAL-TIME EVENTS

Trends

30-DAY HISTORY

Map

DATA FLOWS

In the dashboard

- **Compliance score & risk gauge** — a single 0–100 measure of your posture, updated as findings change.
- **Gateway live feed** — a streaming view of enforcement decisions (masked, blocked, tokenized) with PII types, action and latency.
- **Findings trends** — findings per day over the last 30 days, stacked by severity.
- **Data-flow map** — every detected destination for personal data, colour-coded approved (green) vs. unapproved (red); approve flows you trust.
- **DPDP status checks** — a section-by-section readiness view.

For your platform team

- **Prometheus metrics** — the gateway exposes `/metrics` (requests, PII detections, blocks, LLM calls, latency, uptime) for scraping into Grafana.
- **Health endpoints** — `/healthz` (liveness) and `/readyz` (readiness) on every service for Kubernetes probes.
- **Structured JSON logs** — every service logs with request IDs and tenant context, and never logs raw PII.

OPERATIONAL SIGNALS WORTH WATCHING

Gateway block-rate spikes, sudden bulk-export findings, rising overdue-DSR counts and new unapproved data flows are the leading indicators of a developing incident.

15 Security & multi-tenancy

Security is foundational, not an add-on.

Encryption

- Asset credentials & secrets: **AES-256-GCM** with a **per-tenant key** derived via HKDF from a master key.
- Passwords: bcrypt. API keys: SHA-256 hashed — never stored in clear.
- TLS in transit; HSTS on responses.

Tenant isolation

- **Row-level security** in PostgreSQL scopes every query to one tenant.
- Tenant context is set on each request and each scan.
- One workspace can never see another’s data.

Role-based access control

Role	Capabilities
Owner	Full access, including billing; can delete the workspace.
Admin	Full access except billing.
Analyst	Read + write on findings & policies; cannot manage the team or delete.
Viewer	Read-only across the product (except secrets like API keys).

Authentication & audit

- **RS256 JWTs** with short-lived access tokens and rotating refresh tokens.
- Optional **multi-factor authentication** (TOTP); accounts lock after repeated failed logins.
- Every privileged action is written to an **immutable audit log** retained for years.

DATA SOVEREIGNTY

Defaults target the `ap-south-1` (Mumbai) region, and cross-border transfer policies let you keep regulated data inside India. For complete residency, run the whole platform in your own environment (next section).

16 Deployment options

Run it our way, or entirely your way.

Managed SaaS

We host and operate everything. You connect assets and route gateway traffic. Fastest path to value; nothing to maintain.

Private / self-hosted

Deploy the full stack inside your own VPC or cluster for complete data residency and network isolation.

Docker Compose (single host / SME)

A complete local or single-server stack — control plane, gateway, workers, dashboard, plus PostgreSQL, Redis, ClickHouse and an Nginx TLS terminator. A setup script generates secrets, runs migrations and seeds an admin user.

Kubernetes (Helm)

A production-grade Helm chart deploys each service with sensible defaults:

- Separate deployments per service; one worker deployment per scan queue plus a scheduler.
- Horizontal Pod Autoscaling and PodDisruptionBudgets for the API and gateway.
- cert-manager TLS, ingress routing (/ → dashboard, /api → control plane).
- External Secrets Operator integration, ServiceMonitors and optional NetworkPolicies.

```
helm upgrade --install datasentinel ./helm/datasentinel \
  -n datasentinel --create-namespace \
  -f values.yaml -f values.prod.yaml \
  --set-string image.tag=<release>
```

Baseline requirements

- PostgreSQL 16, Redis 7, ClickHouse 24 (managed or self-hosted).
- A Kubernetes cluster (e.g. Amazon EKS, Mumbai) or a Docker host for Compose.
- A shared 32-byte master encryption key, kept stable and secret across services.

PROTECT THE MASTER KEY

The per-tenant encryption keys are derived from one master key. Keep it identical across services and never rotate it without a migration — rotating it makes previously-encrypted asset credentials unrecoverable.

17 Best practices

A proven path to a strong, low-friction rollout.

1 Start in alert-only mode

Apply policies in alert-only first. Watch the gateway live feed to understand real traffic before you enforce.

2 Connect breadth before depth

Register all major data stores early — even a first scan gives you a map of where personal data concentrates.

3 Use read-only, least-privilege credentials

Every asset connection should use a dedicated read-only principal scoped to only what must be scanned.

4 Triage by severity

Resolve critical Aadhaar/PAN/financial exposures first; mark false positives to keep the signal clean.

5 Route AI traffic through the guard

Make the LLM gateway the only sanctioned path to external models, so PII can never bypass it.

6 Wire alerts into on-call

Send criticals to PagerDuty/Slack so incidents reach a human in minutes, not days.

7 Schedule evidence

Generate a DPDP/audit report on a regular cadence so you always have current evidence on file.

18 Troubleshooting & FAQ

A scan finds nothing / fails

Re-run **Test connection**. Confirm the credential is read-only but has `SELECT / GetObject` on the target schema or bucket, and that network paths/security groups allow the workers to reach the asset.

The gateway returns 400 “X-Upstream-URL is required”

Every gateway request must include the `X-Upstream-URL` header naming the real destination. Add it to your client or egress configuration.

The gateway returns 403 “destination not permitted”

The destination resolved to a cloud-metadata or link-local address, which is blocked for SSRF safety. Use a real external/internal hostname.

A report link doesn’t download

Report files are stored in object storage. Ensure a reports bucket is configured in your deployment; otherwise the platform records a placeholder it can’t serve.

Is my raw data ever stored by DataSentinel?

No. Discovery keeps only metadata and safely-masked samples. The gateway logs hashed/sanitised versions — never raw PII.

Will the gateway add noticeable latency?

Overhead is typically a few milliseconds. If policy can’t be loaded, the gateway fails open so legitimate traffic is never dropped.

Can we keep all data in India?

Yes — use India-region defaults and cross-border policies, or self-host the entire platform for full residency.

A Appendix — Supported PII types

Indian-tuned recognisers with checksum/format validation where applicable.

Identifier	Notes	Typical severity
Aadhaar number	12-digit, Verhoeff checksum validated	Critical
PAN	AAAAA9999A format	Critical
Bank account	Contextual 9–18 digit	Critical
Credit card	Luhn-validated, major schemes	Critical
GSTIN	15-char, checksum validated	High
Passport	Indian passport format	High
Driving licence	State + RTO + number	High
Voter ID (EPIC)	3 letters + 7 digits	High
UPI / VPA	name@bank , all major handles	Medium
IFSC	4 alpha + 0 + 6 alphanumeric	Medium
Phone	+91 / 10-digit Indian mobile	Medium
Email	RFC-style address	Medium
Name / location / date	Detected via NLP	Info
IP / IBAN	Network & international banking	Low

B Appendix — Enforcement actions

Action	Reversible?	Best for
mask	No	Showing partial values (last-4) in UIs/responses.
redact	No	Removing fields that should never leave.
block	—	Hard-stopping prohibited transfers.
tokenize	Yes (vault)	Keeping referential integrity without exposing data.
encrypt	Yes (key)	Protecting fields while preserving recoverability.
hash	No	Pseudonymisation & joins on stable identifiers.
alert	—	Observability without changing traffic.

c Appendix — Glossary & support

Glossary

Term	Meaning
Tenant / workspace	Your isolated organisation account within DataSentinel.
Asset	A connected data resource (bucket, database) that can be scanned.
Finding	A discovered exposure — PII or a misconfiguration — with severity and location.
Policy / rule	The conditions and actions that govern detected personal data.
Gateway event	A logged record of an enforcement decision on live traffic.
Data principal	The individual whom personal data is about.
DSR	Data-subject (principal) rights request under DPDP.
Data flow	A detected destination that personal data travels to.
DPIA	Data-protection impact assessment.

Getting help

Support

Reach the DataSentinel team via your account manager or the in-product help menu. Include your tenant name and the relevant `request_id` from any error for the fastest resolution.

Onboarding

Our team will help you connect your first assets, apply the DPDP starter pack, and stand up the gateway in alert-only mode during onboarding.